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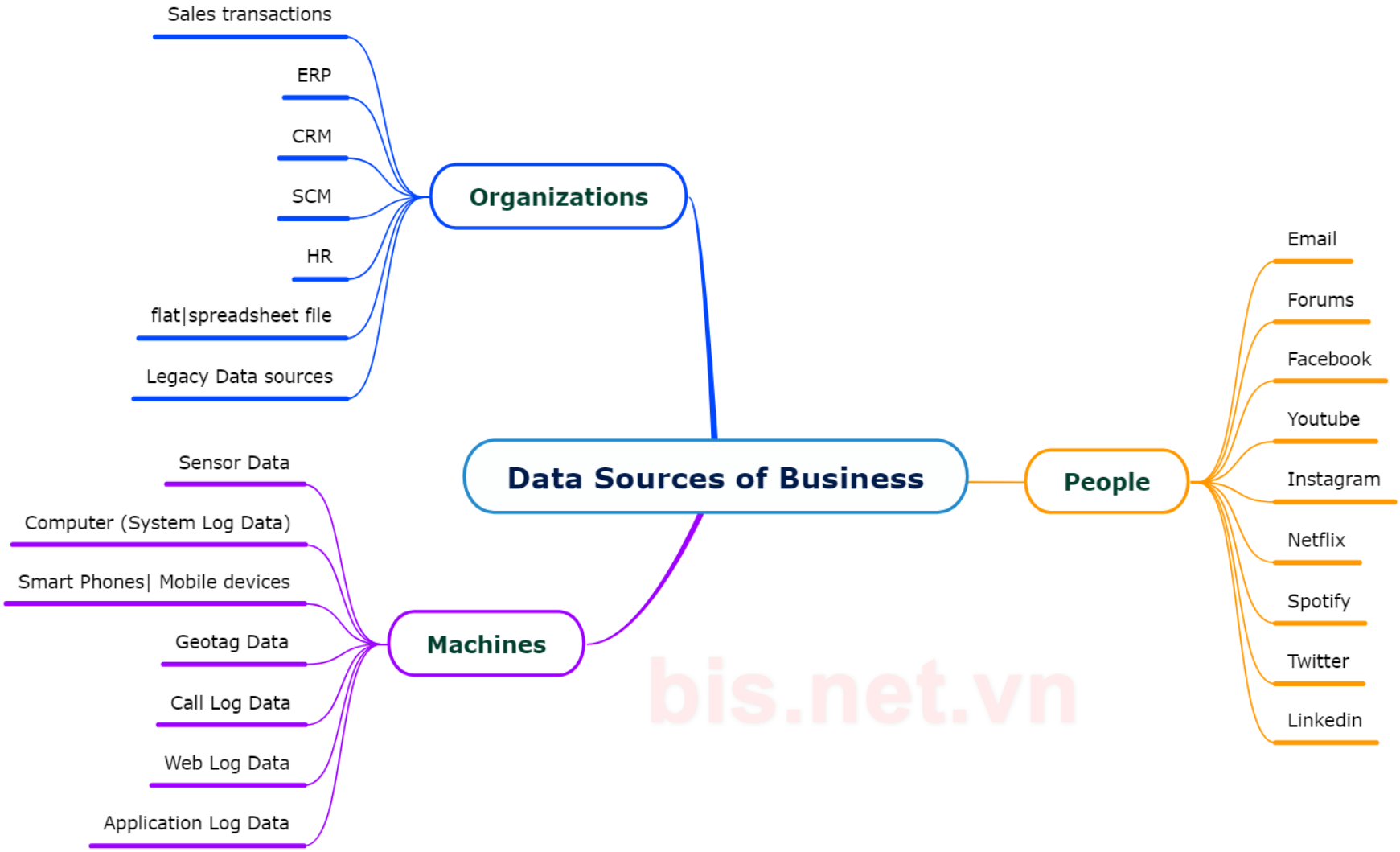
Data & Business Problems

- Digital transformation age
- Big data
- Query and Report
- Improve decision making process
- Access and share information
- Better understand Business
- Friendly with customers & Suppliers
- Develop new products, services, business models

Data Sources



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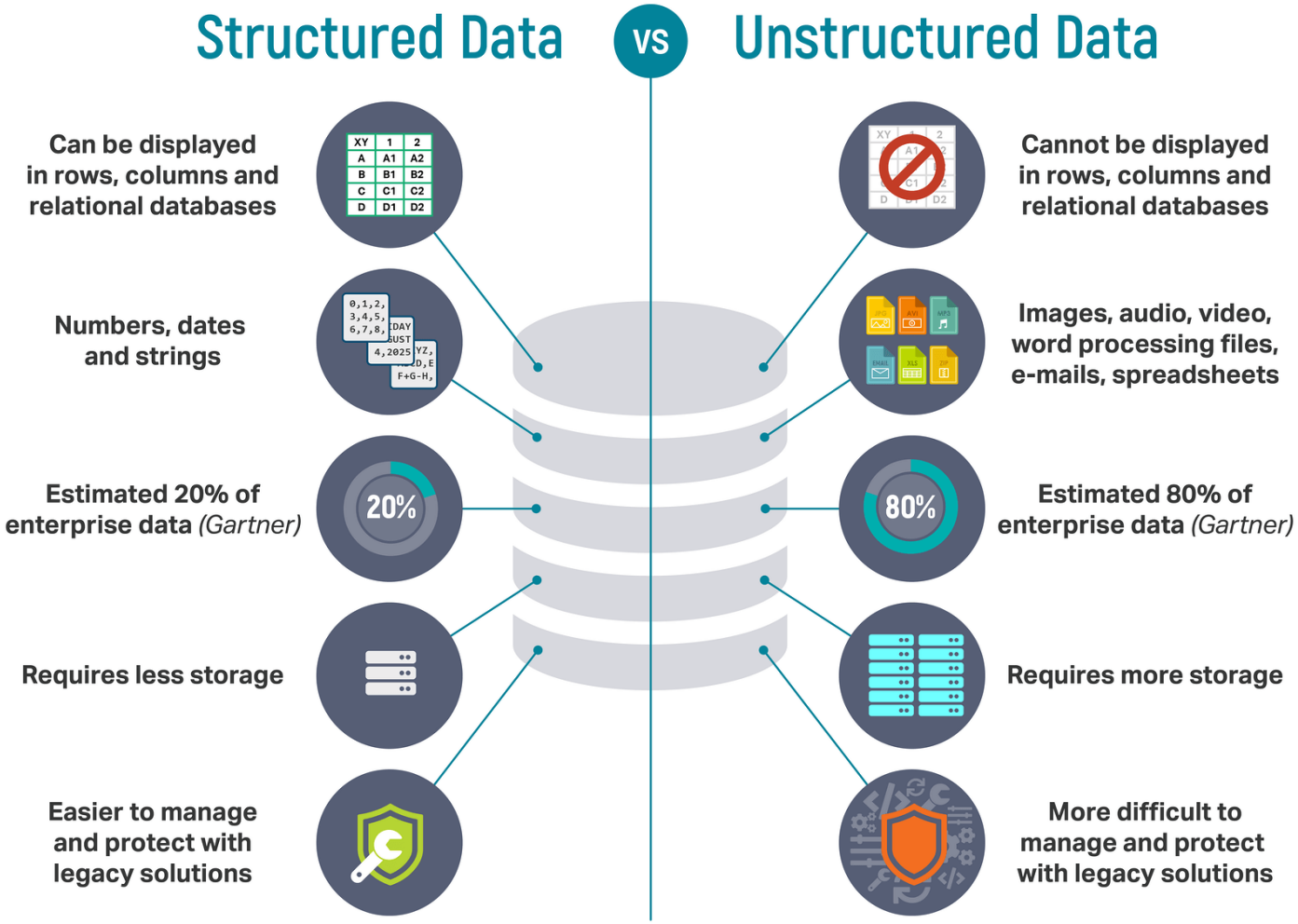
Structured data vs unstructured data



What is Business Intelligence ?

BI is about providing the **right data** at the **right time** to the **right people** so that they can take the **right decisions**.

(*Nic Smith with Microsoft BI Solutions Marketing*)

Business intelligence - BI is the set of techniques and tools for the ***transformation of raw data into meaningful and useful information*** *for business Analysis purposes.*

Why Business Intelligence?

5 Business Purposes

- **Measurement**
- **Analytics**
- **Reporting**
- **Collaboration (Access & sharing data)**
- **Improves the decision Making process (Knowledge Management)**

Why Business Intelligence?

- What happened?=> Report
- What is happening?=> Monitoring
- Why did it happen?=> Analyze
- What will happen?=> Predict
- What do I want to happen?=> Strategy

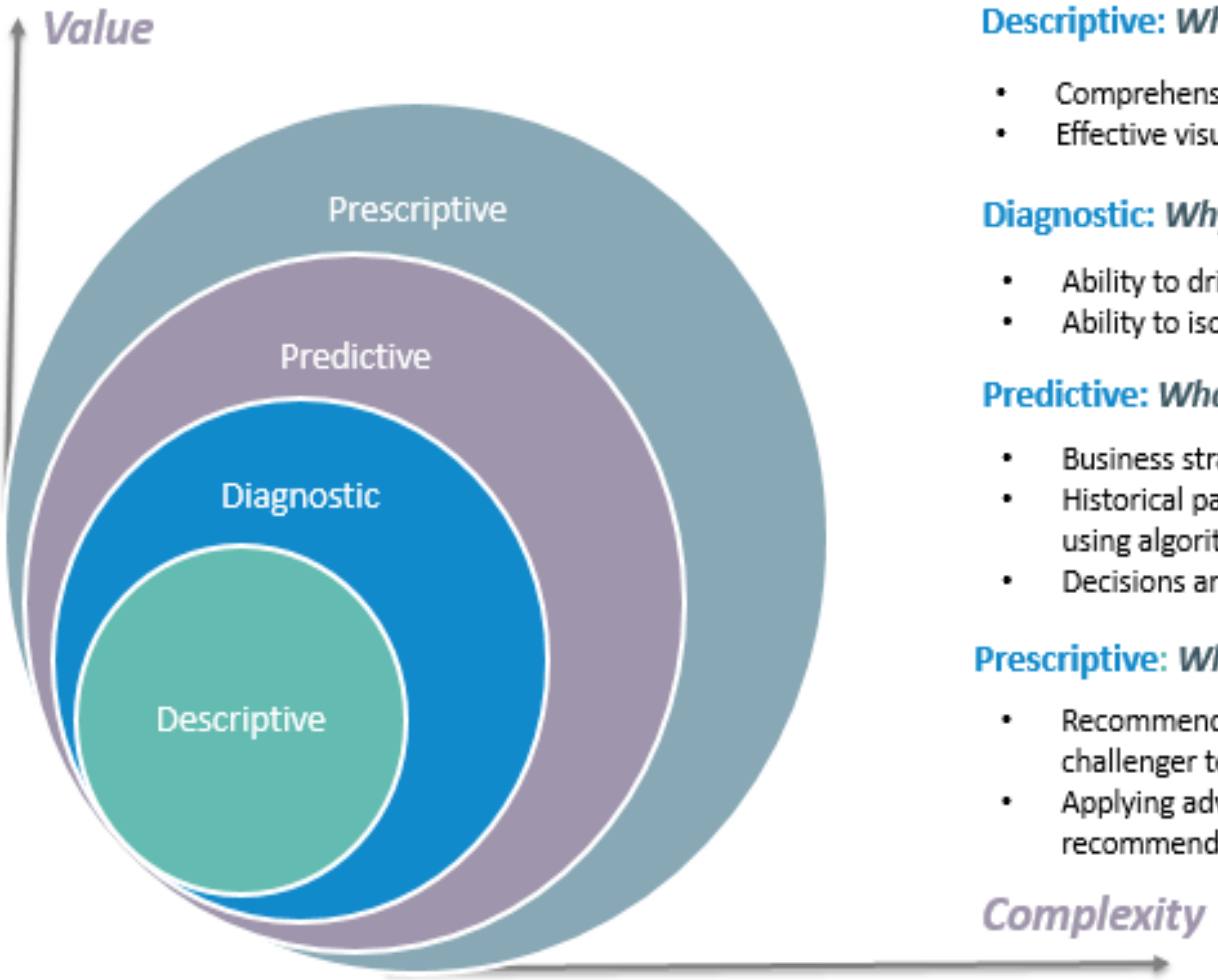
Past

Present

Future

Why Business Intelligence?

4 types of Data Analytics



What is the data telling you?

Descriptive: ***What's happening in my business?***

- Comprehensive, accurate and live data
- Effective visualisation

Diagnostic: ***Why is it happening?***

- Ability to drill down to the root-cause
- Ability to isolate all confounding information

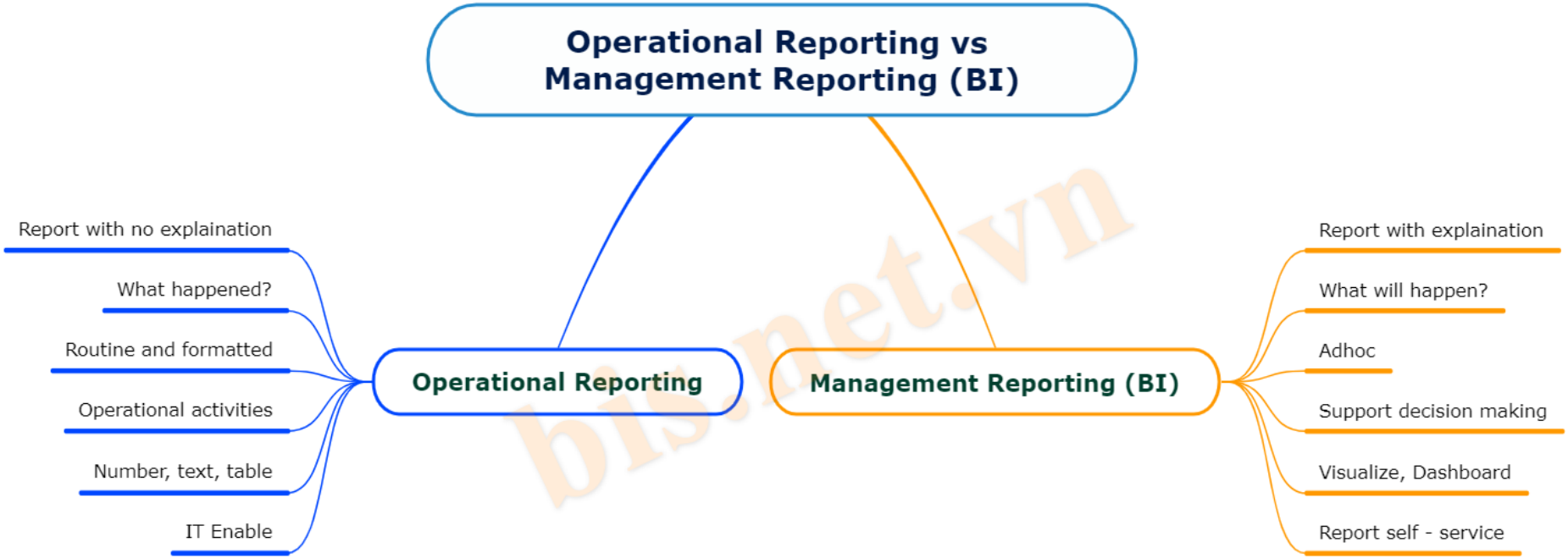
Predictive: ***What's likely to happen?***

- Business strategies have remained fairly consistent over time
- Historical patterns being used to predict specific outcomes using algorithms
- Decisions are automated using algorithms and technology

Prescriptive: ***What do I need to do?***

- Recommended actions and strategies based on champion / challenger testing strategy outcomes
- Applying advanced analytical techniques to make specific recommendations

Operational Reporting vs Management Reporting (BI)



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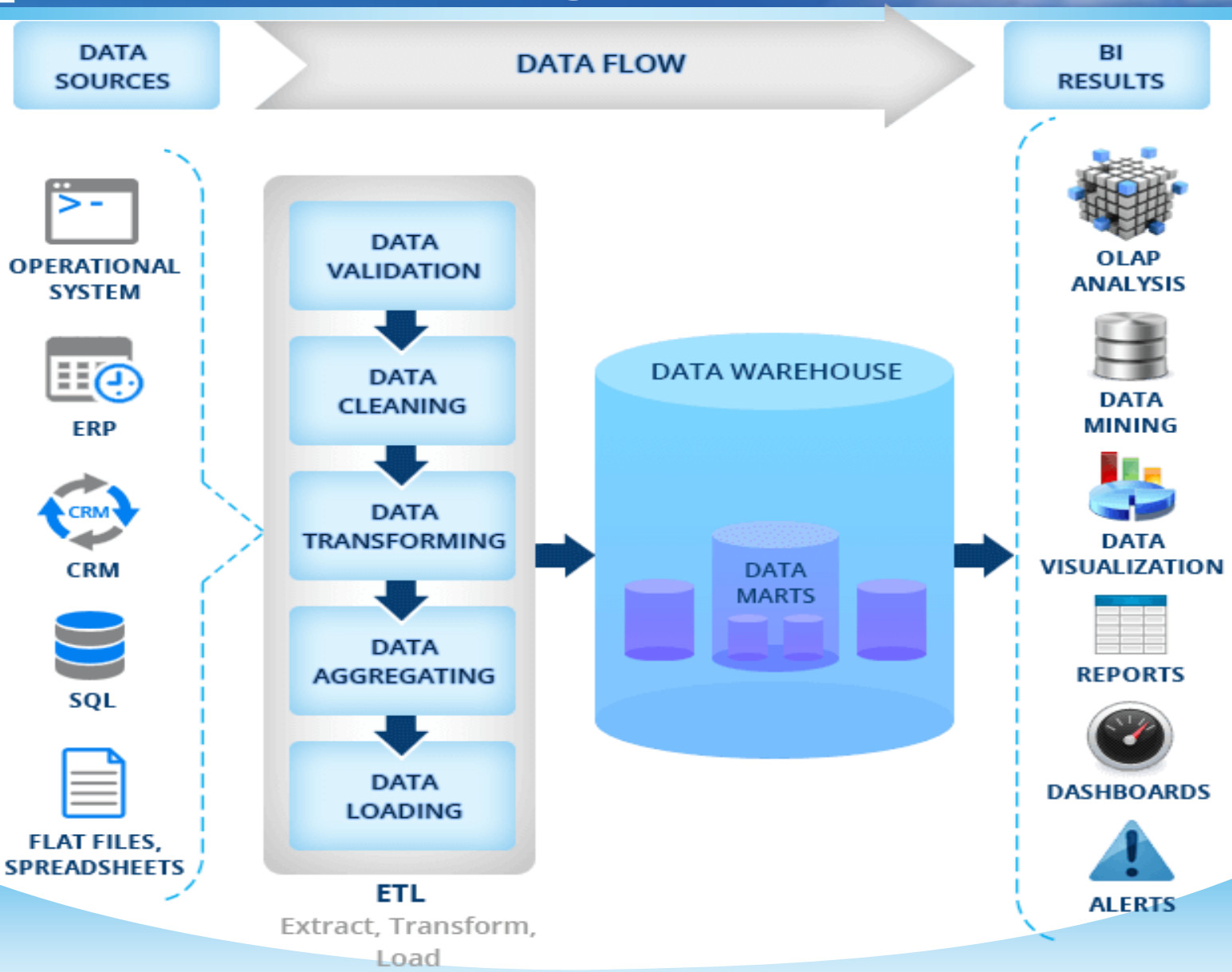
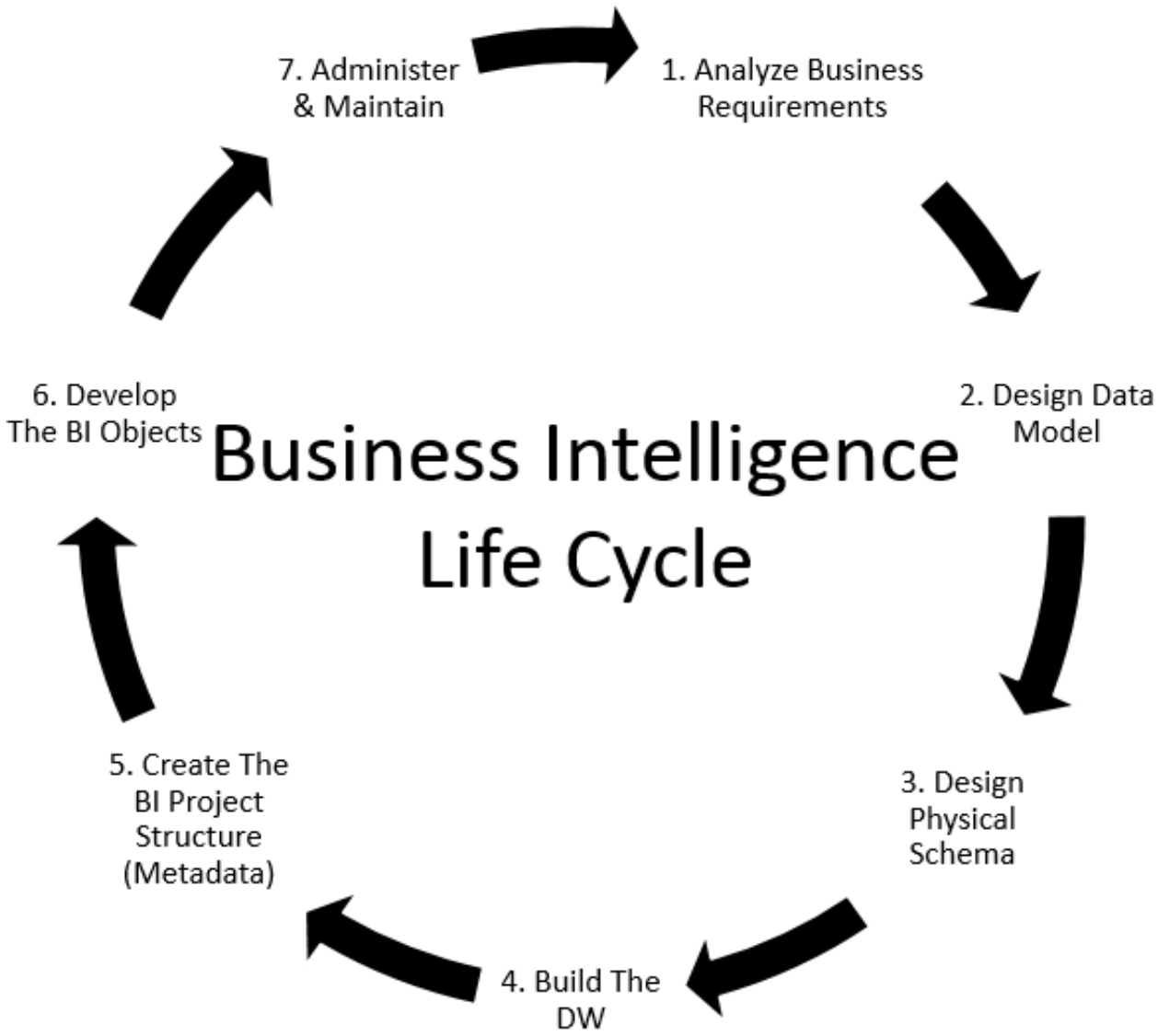
Benefits of BI

Business Intelligence Processes



Business Intelligence Processes



Business Intelligence Activities

- Decision support
- Query and reporting
- Online analytical processing (OLAP)
- Statistical analysis
- Data mining

Technologies support for BI

Data Warehouse: Data Cube, OLAP, MDX

Data Mining:

- Predictive Models
- Data Clustering
- Association Rule
- Data Classification…

Machine Learning

Artificial Intelligence (AI)

IoT (Internet Of Things)

RPA (Robotic Process Automation)

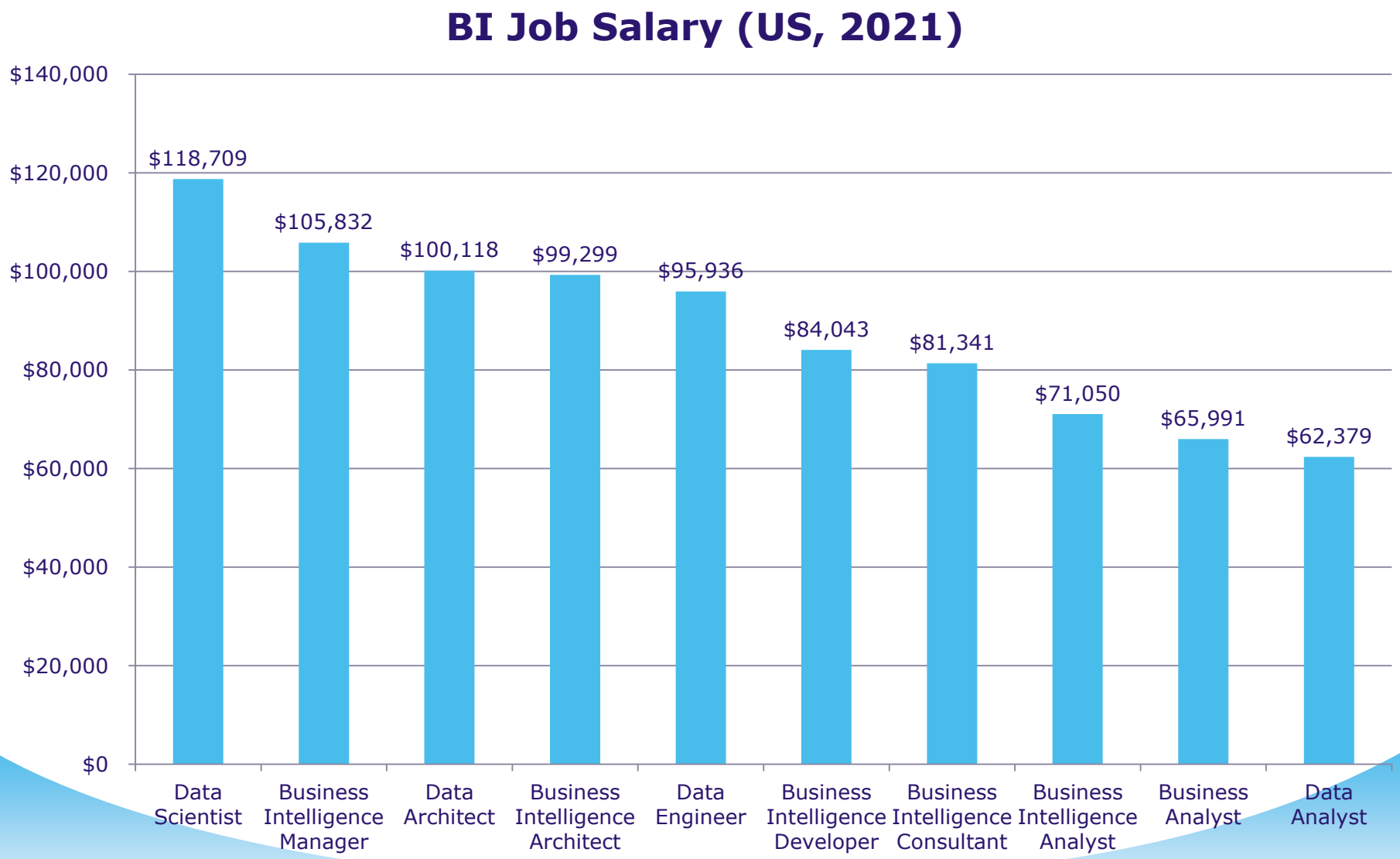
Business Intelligence vision

Microsoft Business Intelligence Vision

Improving organizations by providing business insights to **all** employees leading to better, faster, more relevant decisions

- Advanced Analytics
- Self Service Reporting
- End-User Analysis
- Business Performance Management
- Operational Applications

Business Intelligence Jobs



Business Intelligence Platforms

BI Platforms (2021 Gartner)

Figure 1: Magic Quadrant for Analytics and Business Intelligence Platforms

Source: Gartner (February 2021)

Business Intelligence skill sets

Technical and Practical skills

SQL, Power BI, Tableau, Python, R and Others BI platforms

Strong planning and organization skills

Fast learning

Analytical skills

Soft skills

Collaboration

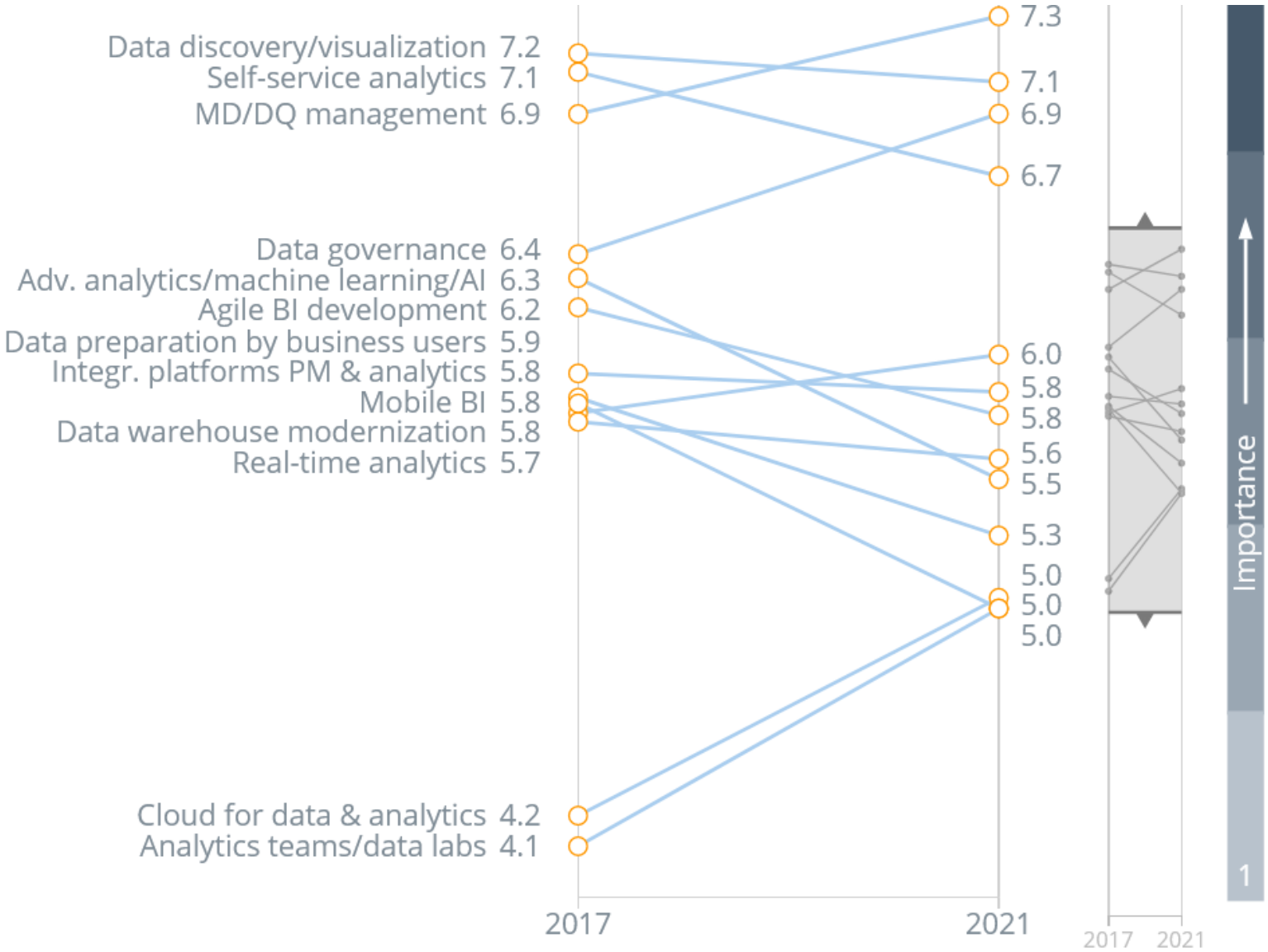
Working independently

Communicate

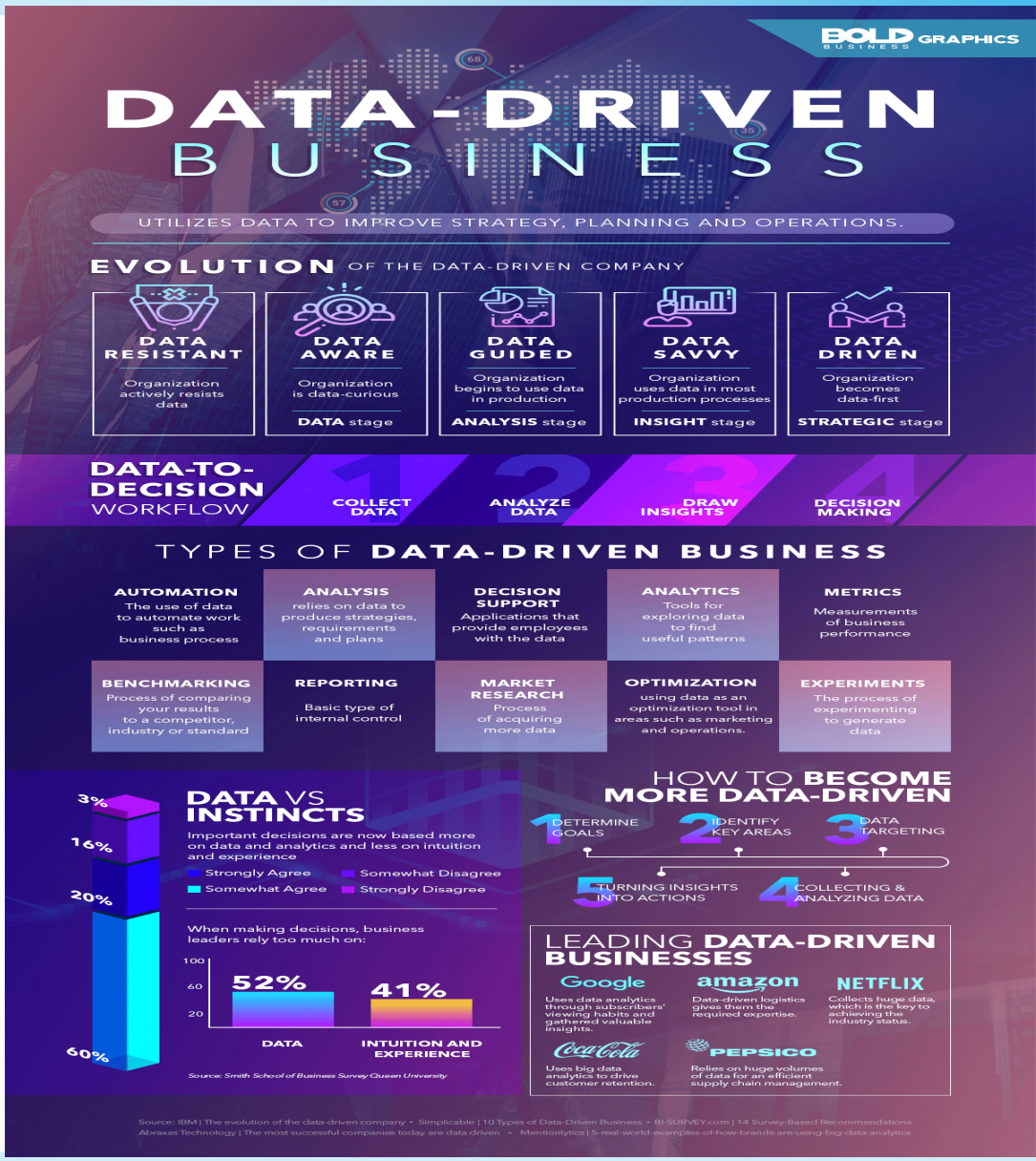
Presentation

Adaptability

Business Intelligence Trends in 2022



Data Driven Organization



Business Intelligence Online Resources

https://www.datacamp.com/

https://www.kaggle.com/

https://www.kdnuggets.com/

https://www.datasciencecentral.com/

https://powerbi.microsoft.com/

https://www.tableau.com/

https://bi-survey.com/top-business-intelligence-trends

https://www.businessintelligence.com/

https://www.analyticsinsight.net/

http://businessintelligence-forum.blogspot.com/

https://www.itjobswatch.co.uk/jobs/uk/business%20intelligence.do

Inspirational Quotes About Data

- “In God we trust, all others bring data”, Edwards Deming
- “Data is the new oil.” , Clive Humby
- “We are drowning in Data but starved for knowledge”, John Naisbitt
- “Torture the data, and it will confess to anything.”, Ronald Coase
- “With data collection, ‘the sooner the better’ is always the best answer.”, Marissa Mayer
- “You can have data without information, but you cannot have information without data.”, Daniel Keys Moran
- “Data beats emotions.”, Sean Rad

Thank you for listening!

For more information, please visit website

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Q&A